



Certificate of Analysis

Vessel / Object: TORM SUBLIME (9854789)
Location: Antwerp / Vesta Terminal Antwerp NV (Belgium)
Job Type: Discharge
Product Grade: Naphtha
Client Reference: Aramco Trading Limited / 1429150

Job No: 632-25-10300
Date Sampled: 08/07/25
Date Tested: 09/07/25
Version: 7 / 09 Jul 2025

Sample 632-25-10300-002
Sample ID, Type & Description SHT 305 AD TUMBL After Discharge

Method	Test	Min	Max	Result	Units	Remarks
ASTM D4052!!	Gravity by Digital Density Meter					
	Density @15°C Top			0.6963	kg/l	
	Density @15°C Upper			0.6964	kg/l	
	Density @15°C Middle			0.6958	kg/l	
	Density @15°C Lower			0.6963	kg/l	
	Density @15°C Bottom			0.6963	kg/l	
	Density @15°C (CALC AVG)			0.6962	kg/l	
	Specific Gravity 15/15°C		0.735	0.6968		
ASTM D5191!!	RVP at 37.8 °C		12.5	9.43	psi	
ASTM D156	Saybolt Color	20		+30		
ASTM D6839*	Hydrocarbon Types, Oxygenated Compounds and Benzene by GC					
	Olefins		1.0	0.3	%vol	
	Aromatics			7.3	%vol	
ASTM D6839	Hydrocarbon Types, Oxygenated Compounds and Benzene by GC					
	Naphthenes			25.9	%vol	
	Paraffins	65		66.4	%vol	
	N-Paraffins			28.6	%vol	
	Iso-Paraffins			37.8	%vol	
	N+A			33.2	%vol	
ASTM D5453!!	Total Sulfur Content		500	148	mg/kg	
ASTM D3606 Proc. A	Benzene			2.82	%vol	
UOP 163	Hydrogen Sulfide		10	< 1	ppm	
AA/Graphite Furnace	Lead		50	< 20	ppbw	
UOP 779	Organic Chlorides		3	< 1.0	ppmw	
ASTM D7423	Oxygenates by FID GC					
	Total Oxygenates		100	< 10	mg/kg	
	Methyl Tert-Butyl Ether (MTBE)		50	< 10	mg/kg	
UOP 938	Mercury		5	0.690	ppbw	
ASTM D3227	Mercaptan Sulfur			59	ppmw	
ASTM D2699	Research Octane Number by Bracketing-Equilibrium (Uncorrected)			66		
ASTM D2700	Motor Octane Number by Bracketing-Equilibrium (Uncorrected)			64		
GC-MS	Carbon Disulfide (CS2)			1	mg/kg	
ASTM D86!!	Distillation Evaporated					
	IBP	30		32.6	°C	
	5%			43.3	°C	
	10%			49.5	°C	



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 SHT 305 AD TUMBL After Discharge

<u>Method</u>	<u>Test</u>	<u>Min</u>	<u>Max</u>	<u>Result</u>	<u>Units</u>	<u>Remarks</u>
ASTM D86!!	Distillation Evaporated					
	20%			55.5	°C	
	30%			61.4	°C	
	40%			67.7	°C	
	50%			75.2	°C	
	60%			84.0	°C	
	70%			94.5	°C	
	80%			108.3	°C	
	90%			126.7	°C	
	95%			140.2	°C	
	FBP		180	156.6	°C	
	Residue			1.2	%vol	
	Loss			2.9	%vol	

Notes: *Method Modification: Result is reported outside (lower/higher) the scope of the method

Simon Ryckaert
 Laboratory Planner

!! Test Method performed and reported within the scope of EN/ISO 17025:2017 accreditation (BELAC 735-TEST).
 All methods are performed according to the latest version available unless specified otherwise in this report.
 Measurement uncertainty (applicable to the evaluation of test results), while not included in this report, can be made available upon first request.
 The results in this report relate only to the items tested.